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## **Imitation Learning in Robotics**

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## Abstract

Imitation Learning (IL) has emerged as an efficient approach for end-to-end robotic skill acquisition, offering an alternative to manually programmed controllers and reward-driven reinforcement learning. This review traces IL’s development from early behavioural cloning and inverse reinforcement learning methods to more modern generative, diffusion and transformer model strategies. In this review common issues such as covariate shift, causal misidentification and multi-modality while exploring various solutions, including interactive, offline and implicit methods are discussed. Additionally, covering more recent advancements in inverse reinforcement learning and adversarial approaches as well as diffusion policies and behavioural transformers, capable of modeling complex action distributions. This survey also digs into imitation from observation, where agents learn effectively from video demonstrations without explicit action data. After giving a thorough overview about the algorithmic of imitation learning, recent applications of IL on various robotic systems, showing practical capabilities and potential of IL in diverse areas of robotics, such as manipulation, autonomous driving, legged locomotion and aerial navigation are investigated.

## 1 Introduction

The field of robotics experienced great advances in recent years, making it possible to perform many tasks, usually handled by humans automatically. However, robots still face difficulties in performing detailed, context-specific actions with the same flexibility and adaptability as humans. Humans are naturally skilled at adjusting quickly to changing environments, something robotic systems typically struggle with [1]. Closing this performance gap between humans and robots or even surpassing human skills is still an ongoing goal in robotics research.

Imitation Learning (IL) is one method used to deal with this, by teaching robots to perform tasks based on human demonstrations. This allows robots to learn skills without manually programming every single action [2]. Furthermore, human demonstrations could also be thought of an initial heuristic, that robot can refine and enhance, potentially surpassing the original human performance. IL has evolved significantly over time, starting with early demonstrations like ALVINN by Pomerlau et al. [3], followed by important methods such as behavioural cloning (BC) and inverse reinforcement learning (IRL), while still continuing to recent innovations.

Compared to reinforcement learning (RL), which requires designing reward systems and often needs large amount of data to learn effectively, the idea behind IL is to benefit from expert demonstrations, making it more efficient and easier to apply [4]. This advantage is especially beneficial in situations, where safety is critical and rewards for correct actions are sparse [5].

Despite these benefits, IL still faces several important challenges. These include covariate shift, which occurs when differences between training and real-

world data cause performance issues [5], causal misidentification, where the IL algorithm incorrectly identifies the reasons behind certain actions [6] and multi-modality, which happens, when multiple correct actions exist for the same situation, making learning and decision making complicated [7].

Recent advances in IL such as diffusion policies [8] and behavioural transformers [9] have provided strong methods to address these challenges. These new techniques have significantly improved IL performance for robotics systems in tasks such as robotic manipulation [10] autonomous driving [11] and legged locomotion [12], demonstrating ILs practical potential.

However, there are still many open research questions and opportunities for further exploration. Important areas for future studies include IL policy transfer to new environments, creating better simulation environments for decreasing covariate shift and making IL systems scalable for real-world and large-scale use as well as improving their transparency.

This review essay covers the key methods and recent advances in IL within robotics. By summarizing current techniques and outlining challenges, this review aims to provide a clear picture of IL research trends as well as identifying promising areas for future research and development.

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## 2 Imitation Learning

### 2.1 Behavioural cloning

Behavioural Cloning (BC) is the first class of methods we discuss for solving an imitation learning (IL) task. Given expert demonstrations (trajectories) consisting of state-action pairs  $\mathcal{E} = \{(s_i, a_i)\}_{i=0}^N$ , the goal is to find a function that mimics the experts underlying policy. BC treats policy learning as a supervised learning problem. When a function is trained using expert demonstrations, it learns to map previously unseen observations to expert actions.

BC is conceptually simple and straightforward to implement, since it uses standard supervised learning techniques, which is a well-studied field in machine learning [13]. It avoids explicit reward function design and operates well in tasks, where large numbers of high-quality demonstrations are available. Moreover, BC does not require knowledge of the system dynamics because its training process relies solely on expert demonstrations.

Despite these advantages, BC has several limitations. One is that the model’s performance depends on the quality of expert demonstrations, also known as ‘ceiling effect’ [14]. Another limitation is the model’s lack of interpretability compared to other IL methods that learn underlying reward functions. As a result, BC often suffers from ‘causal misidentification’, where the model does not take into account the true causal relationships within its environment. Additionally, BC requires sufficiently large and diverse datasets to ensure generalization. Furthermore, a critical drawback of BC is its sensitivity to covariate shift, described in Section 2.2. Later, we will discuss algorithms, which struggles with

multi-modality, which occurs when there are multiple valid ways to act, given the same state (see figure 1). Since expert data often contains several plausible actions an agent that combines them into an average action might fail the task.

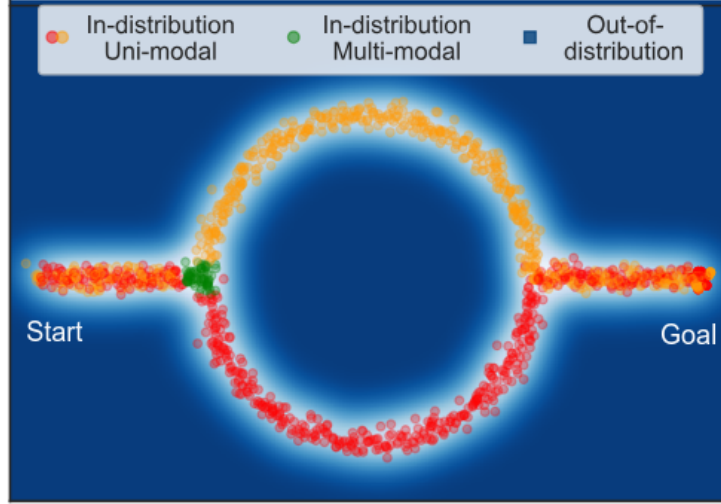


Figure 1: To reach the goal state from the start, a clockwise and a counterclockwise direction are possible. While the orange and red dots visualize a uni-modal path, the green dots represent the multi modal region, where multiple diverging paths can be taken

## 2.2 Covariate Shift

In machine learning we usually assume, that both training and test data are independently and identically distributed (i.i.d) from the same underlying distribution.

$$(x, y) \sim P_{train}(X, Y) = P_{test}(X, Y)$$

Covariate shift arises when the distribution of input features changes between training and testing, while the conditional relationship  $P(Y|X)$  between inputs and outputs remains the same.

$$P_{train}(X) \neq P_{test}, \quad P_{train}(Y|X) = P_{test}(Y|X)$$

This is particularly problematic in BC. Although the agent learns a policy  $\pi(a|s)$  that mimics the expert's action distribution  $\pi_E(a|s)$  during training, the testing generates a different trajectory of state-action pairs. Even small deviations between the expert's and the agent's actions, can lead the agent into unexplored states, where the agent might perform poorly.



If the agent makes a small error  $\epsilon$  at each timestep, these errors accumulate as the agent follows its policy, continuously diverging from those in the training data. According to [5], this accumulation of errors grows quadratically with the length of the trajectory  $T$ , yielding  $\epsilon \in \mathcal{O}(T^2)$ . Therefore, BC policies that perform well under training conditions can fail during execution.

### 2.2.1 Interactive IL

Covariate shift poses a risk, especially in safety-critical environments, where recovering from mistakes is crucial. One research direction that addresses this problem is interactive IL, where the algorithms assume access to an online expert who can provide actions in uncertain states during training.

Dataset Aggregation (DAGger) [5] is one of the first algorithms in this category. It addresses the covariate shift by collecting data from the policy’s own trajectories. In each iteration, the current policy is deployed in the environment, while an expert provides actions for newly encountered states. These new state-action pairs are then combined with previously collected data to retrain the policy. This data aggregation step promotes better recovery from unseen states because the policy explicitly learns to handle previously critical states. As a result, the error accumulation per trajectory is reduced from quadratic to linear complexity  $\epsilon \in \mathcal{O}(T)$  [5].

However, a major drawback of DAGger are the frequent expert interventions, which are required to provide corrective actions. If the expert is human, this can lead to a high cognitive burden with the risk of inaccurate feedback, negatively affecting the training process.

To address this issue, various interactive IL extensions have been invented, roughly classified as ‘human-gated’ and ‘robot-gated’ approaches.

### 2.2.2 Human-Gated Methods

In human-gated methods, the expert decides when to provide corrections during executing the agent’s policy execution.

For instance, Human-gated HG-DAGger [15] extends DAGger by intervening only when the agent’s state reaches an unsafe state (see figure 2). The expert corrects the agent’s trajectory until it reaches a safe region, where the agent resumes its policy execution. However, in the worst case, this method could still lead to frequent human intervention.

To minimize this burden of expert intervention [16] introduced a method that defines an intervention cost, incentivizing the agent to reduce frequent expert corrections.

Similarly, [17] proposed a human-in-the-loop technique called contrastive policy learning from interactive corrections (CLIC), which constructs a desired action space from human feedback rather than cloning the expert’s actions. This technique increases robustness during learning and prevents overfitting to noisy or suboptimal demonstrations.



Figure 2: During an autonomous driving task (left), the human expert (right) has to surveil the cars actions and intervene in undesired states

Other work by [18] includes human gaze data into the training process to steer the model’s attention towards regions that people focus on while recording their demonstrations. (see Figure 3).

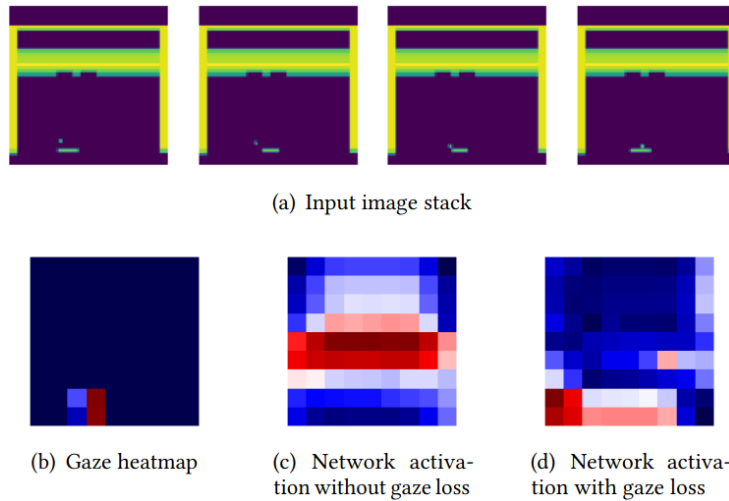


Figure 3: During demonstration collection by playing the Atari Breakout game (a) the gaze heatmap (b) is recorded. Incorporating this heatmap, switches the network activation, shown in (c) and (d) to the point of interest

Although human-gated approaches can reduce the expert’s interventions, the expert has to maintain constant attention while monitoring the training process.

### 2.2.3 Robot-Gated Methods

Having a human maintaining continuous attention can be exhaustive. Robot-gated IL addresses this problem by actively querying the human expert on demand. One of such extensions of DAgger is SafeDagger [19]. SafeDagger

uses a safety policy  $\pi_{safe}$  to predict potential policy deviations from the expert’s actions in a given state. When  $\pi_{safe}$  detects such a risk, it requests the expert’s intervention.

LazyDagger [20] extends SafeDagger further by reducing the ‘context switches’ between the agent and the human expert. These switches are a burden for the expert. To minimize the burden, they add Gaussian noise to the supervisor’s actions during interventions to increase state-space exploration.

ThriftyDagger [21] introduces two criteria to determine when a robot should request the expert’s help: (1) the agent’s current state is significantly out of distribution and (2) there is a high risk of failing the task in the current state. Additionally, the expert can also preset a budget of context switches, further limiting the burden.

EnsembleDagger [22] approximates a Gaussian process by using an ensemble of neural networks that estimate policy uncertainty by a ‘policy disagreement’. Similarly to ThriftyDagger, control is only handed to the expert, when the expert’s state is far from the training distribution or if the ensemble’s variance surpasses a specified threshold. This approach balances safety with exploration by focusing on new and important states.

Recently, Lee et al. [23] introduced Diff-Dagger, which uses diffusion policy training objectives (see Section 2.5) to better estimate uncertainty in robotic manipulation tasks. This approach addresses a key limitation in robot-gated Dagger approaches for multi-modal tasks.

#### 2.2.4 Offline BC

The previous discussed BC methods always rely on an online expert. If no online expert is available, the agent must learn solely from prerecorded expert trajectories, which often lacks size or sufficient coverage of the state space. DART (Disturbances for Augmenting Robot Trajectories) [24] addresses this by injecting noise during data collection to force the expert into recovery situations. This approach leads to a more diverse dataset, in which the expert not only showcases optimal trajectories, but also corrective behaviour. As a result, this method achieves performance, comparable to online expert methods like Dagger while maintaining the computational efficiency of supervised learning tasks.

Instead of injecting noise during data collection, Yadgaroff et al. [25] investigated various data augmentation techniques and their influence on experts generalization capabilities, after expert data collection. Their findings indicate that combinations of multiple augmentations can improve the agents performance by up to 80% compared to unmodified data.

#### 2.2.5 Implicit BC

In contrast to classical BC, which directly maps observations to expert actions, implicit BC (IBC) models the relationship between observations, actions and expert behaviour implicitly. Rather than learning a direct mapping, IBC trains an energy function  $E(s, a)$  that assigns low values to state-action pairs that

match expert behaviour and high values otherwise. During inference the agent selects the action

$$a^* = \underset{a}{\operatorname{argmin}} E(s, a)$$

returning the most expert-like action [26].

Compare to classical BC, IBC takes longer to train and is computationally more demanding. However it has better performance on multi-modal tasks, higher sample efficiency and tends to generalize better [26].

### 2.2.6 Targeting Causal Misidentification

As discussed in Section 2.1, ‘causal misidentification’ occurs when BC agents fail to learn the causal factors driving expert’s actions but learning correlations instead. DeHaan et al. [6] addressed this issue by extending the agent’s policy input by a ‘causal graph’ structure. This graph represents hypotheses about the true variables that influence the expert’s actions. During execution, the agent can sample from different causal graphs, while still being able to query the expert in confusing situations.

An interesting example demonstrating causal misidentification is the ”copy-cat problem” mentioned by [27]. When an agent policy is trained on multiple past observations, it can learn to predict the next action by just copying its own previous actions, rather than responding to environmental cues. Surprisingly this problem lead to a policy, where an agent trained on a single past observation, outperformed an agent observing multiple past observations. To solve this, Wen et al. [27] proposed an adversarial approach, involving a target-conditioned adversary (TCA). This adversary attempts to predict the previous action using the encoded state representation and the current action of the agent. Meanwhile, the encoder trains to prevent this prediction while still maintaining useful information for the agent’s policy. As a result, the agent can no longer simply rely on encoded information of past actions, but has to learn a causal relationship between environment states and appropriate actions.

## 2.3 Inverse reinforcement learning

The methods reviewed so far improved the supervised learning approach behind BC. IRL on the other hand uses a different approach. From expert demonstrations it first infers a reward function, that could have generated the experts behaviour and then solves the resulting RL problem. Because the agent repeatedly interacts with the environment to maximize the cumulative reward, IRL offers better recovery from off-distribution states and is therefore less sensitive to covariate shift than BC.

However, IRL introduces two major difficulties. A single optimal policy can correspond to infinitely many reward functions [28]. For example, for any positive integer  $c \in \mathcal{N}_{>0}$

$$R_c(s, a) = c * R'(s, a), c \in \mathcal{N}^+$$

with a fixed non-zero  $R'$ , the same policy  $\pi^*$  remains optimal. This problem is also known as ‘reward ambiguity’. The second challenge is the computationally cost of IRL algorithms, because most of them alternate between estimating a reward and solving an RL sub-problem, which is expensive [29].

Current IRL methods fall into two broad categories, where the first category describes the iterative learning process by repeatedly interacting with the environment (see Section 2.3.5), while the second category describes fixed-reward schemes that define a distance-based reward once and keep it unchanged during training (see Section 2.3.5).

### 2.3.1 Maximum Margin Methods

Maximum-margin IRL addresses reward ambiguity by defining a strictly higher return for expert trajectories, than any alternative, by a user-specified margin. Formally, with the reward  $R_\omega(s, a) = \omega^\top f(s, a)$  this can be described by the following formulation:

$$\omega^\top \mathcal{E}(\pi^*) \geq \omega^\top \mathcal{E}(\pi) + L(\pi), \quad \forall \pi \in \Pi$$

where:

- $\omega$  represents the weight vector that parametrizes the reward function
- $\mathcal{E}(\pi)$  represents the feature expectation of a policy
- $L(\cdot)$  represents the loss function, measuring dissimilarity between policy  $\pi$  and expert policy  $\pi^*$
- $\Pi$  represents the set of all possible policies

Ratliff et al. [30] introduces maximum margin planning (MMP) that formulates the imitation learning problem as a structured maximum margin optimization problem. Their optimization simultaneously selects  $\omega$  and the induced policy so that demonstrated trajectories are separated from sub-optimal ones by the largest possible margin.

### 2.3.2 Maximum Entropy Methods

An alternative is to choose among all reward functions that match the experts policy the one that returns the highest entropy of their trajectory distribution. Formally, the maximum-entropy distribution over trajectories  $\zeta = \{(s_0, a_0), \dots (s_T, a_T)\}$  is

$$P(\zeta|\omega) = \frac{1}{Z(\omega)} e^{\omega^\top f_\zeta}, \quad f_\zeta := \sum_{(s,a) \in \zeta} f(s, a)$$

where

- $\omega$  represents the weight vector that parametrizes the reward function

- $Z$  represents the partition function, normalizing the distribution
- $f_\zeta$  represents the sum of features along the trajectory (empirical feature count)

The goal is to maximize

$$H(P) = - \sum_{\zeta} P(\zeta|\omega) \log P(\zeta|\omega)$$

where

- $H$  entropy of distribution  $P$

with subject to matching the empirical feature count:

$$E_{\zeta \sim P(\zeta|\omega)}[f_\zeta] = \tilde{f} := \frac{1}{m} \sum_{i=1}^m f_{\tilde{\zeta}_i}$$

where

- $\tilde{\zeta}_i$  represents expert demonstration trajectories
- $\tilde{f}$  is the empirical feature count from expert demonstrations

Ziebart et al. [31] introduced this principle into their maximum entropy IRL (MaxEntIRL) showing improved robustness to sub-optimal demonstrations compared with MMP.

Subsequently Wulfmeier et al. [32] extended MaxEntIRL by a convolutional neural network (CNN) for automatic feature extraction replacing manual feature extraction and non-linear reward functions, making the approach more suitable for complex and high dimensional domains.

Snowell et al. [33] further extended MaxEntIRL by an exact gradient computation for stochastic MDPs using KL-divergence, replacing the approximate heuristical approach used in MaxEntIRL.

### 2.3.3 Bayesian methods

Bayesian IRL treats the reward as a random variable and infers its posterior given demonstrations:

$$P(R|O_X) = \frac{P(O_X|R)P(R)}{P(O_X)}, \quad O_X = \{(s_1, a_1), \dots, (s_T, a_T)\}$$

where

- $R$  is the reward function
- $P(R)$  is the prior distribution over reward functions

- $P(O_X|R)$  is the likelihood of observing the experts behaviour given reward function  $R$ .

The reward is then estimated as the expected value under this posterior:

$$\hat{R} = E_{R \sim P(R|O_X)}[R].$$

This enables the agent to reason about uncertainty and handles reward ambiguity.

Ramachandran et al. [34] used the Boltzmann distribution in their BIRL algorithm for modeling the likelihood and samples the posterior via Markov-chain-Monte-Carlo (MCMC). Because each sample required solving the MDP, BIRL scaled poorly.

Follow up work by Mandyam et al. [35] proposed KD-BIRL (Kernel Density BIRL), replacing the Boltzmann likelihood with conditional kernel density estimates between state-action pairs, reducing sample complexity in high-dimensional space.

Brown et al. [36] introduced bayesian reward extrapolation (B-REX), making bayesian reasoning practical for high-dimensional tasks, by using deep neural networks with a successor feature representation (factoring the value function in to dynamics and reward parts). This enables bypassing repeated MDP solutions. When applying this method to Atari games better-than-expert policies from suboptimal demonstrations could be learned.

### 2.3.4 Adversarial IL Methods

Adversarial IL (AIL) addresses both reward ambiguity and computational cost by training a policy and a reward model simultaneously. Inspired by generative adversarial networks (GANs) [37], a discriminator  $D_\phi$  learns to distinguish expert from agent state-action pairs, while the policy  $\pi_\theta$  is updated to fool the discriminator. The reward is then given by the discriminators discrimination confusion as  $R(s, a) = -\log D_\phi(s, a)$ .

Ho et al. realized this in their generative adversarial IL (GAIL) [4] approach by using two identical multi-layer perceptrons (MLPs) for discriminator and agent, while using trust region policy optimization (TRPO) for policy updates. GAIL mitigates the problem of reward ambiguity by avoiding an explicit reward function reconstruction. Instead the policy is learned directly from demonstrations.

Adversarial Inverse RL (AIRL) [38] modifies GAILs discriminator such that it is forced to separate a task reward, capturing the true objective and a value-based shaping term, containing information about the underlying environment. This makes AIRL adaptable to changing environment dynamics.

Wasserstein AIL (WAIL) from Xiao et al [39] address common GAN problems such as mode collapse [40], oscillations [41] and vanishing gradients [42]. It replaces the Jensen-Shannon divergence, measuring the similarity between agent and expert state-action distributions by the Wasserstein distance. By

deriving the reward from the corresponding Kantorovich potential, the reward becomes more meaningful throughout training, leading to more stable training.

Kostrikov et al. [43] found that the reward function in GAIL is implicitly biased, which can introduce undesirable prior assumptions. This incentivizes the policy for unnecessary exploration or terminate the episode prematurely. Additionally, GAIL is sample inefficient, requiring a lot of environment interactions. Their discriminator-actor-critic (DAC) method replaced GAILs on-policy TRPO with off-policy TD3, added a reward buffer and explicitly assigning rewards for absorbing (final) states. Together these changes decreases the number of required environment interactions and solved the reward bias.

### 2.3.5 Constant Reward Functions

Instead of learning a reward online, like discussed in sections 2.3.1 - 2.3.4, a predefined fixed reward that measures how close the agent stays on the expert’s state-action distribution (also called support). The RL problem then only has to be solved once with this reward.

Dadashi et al. [44] introduced primal Wasserstein imitation learning (PWIL). They use the Wasserstein distance between expert and agent state-action distribution as a reward signal, without the need of online training, requiring minimal parameter tuning. Their method was successfully learned Humanoid policies from a few sub sampled demonstration.

Wang et al. [45] came up with a method called random expert distillation (RED), where they estimate this support of the experts policy using RND (random network distillation). Here the reward function consists of a network, that returns a high reward for being "close" to the experts support and a low reward for visiting novel states. In addition the authors also propose the use of an autoencoder network instead of RND, where the reward is based on the reconstruction error of state-action pairs.

Reddy et al. [46] proposed SQIL (soft Q IL). SQIL associates expert demonstrations with a reward of 1 and agent transitions with a reward of 0, collecting them in a replay buffer. Using soft Q-learning by sampling balanced batches (50% expert demonstrations, 50% new experiences) the agent’s policy is trained.

Finally, the disagreement-regularized IL (DRIL) from Brantley et al. [47] trains an ensembles of policies on expert demonstrations. The reward is defined as the ensemble disagreement (ensemble variance), which is low on visited states and high elsewhere.

## 2.4 Imitation from Observation

Most imitation learning algorithms reviewed so far assume access to expert data in the form of state-action pairs. From a human learning perspective, this sounds rather counter-intuitive. Usually humans infer motor commands by watching others, rather than being provided their joint torques. Despite never seeing the expert’s controls, humans generalize across many tasks and environments [48], [49].



The same observation motivates imitation from observation (IfO). Collecting large, labeled state-action datasets is expensive, time consuming to collect and often from simulators that do not transfer well to the real-world domains. On the contrary, unlabeled demonstrations in form video data all over the internet or unstructured sensor data is plentiful. IfO methods therefore aim to learn policies directly from an observation trajectory  $\mathcal{O} = \{O_1, \dots, O_T\}$  without ever seeing the expert’s actions.

Learning from observations introduces several difficulties. First, imperfect time-synchronization causes temporal misalignment, where the expert often performs the task faster, slower or out-of-phase with the agent, making frame-level matching unreliable [50]. Furthermore, because of differences in morphology, kinematics and viewpoints lead to a human demonstrator’s motion cannot be mapped directly onto a robot’s actuators. This problem is known as cross-embodiment correspondence [51]. Similarly, demonstrations usually originate from distinct simulators, cameras or physical settings, producing a domain-mismatch and therefore a covariate shift between the training data and the agent’s environment [52]. Also, since raw videos suffer from occlusion, background clutter and sensor noise, agent’s must operate under partial observability and still extract relevant cues. IfO algorithms must therefore overcome misalignment, cross-embodiment gaps, domain mismatch and remain robust under partial observability, while simultaneously inferring latent actions of the experts.

Torabi et al. [53] introduced BCO (Behavioural Cloning from Observation). In their approach they split the training into a pre-demonstration and post-demonstration phase. In the pre-demonstration, the agent executes a random policy to gather  $(s, s, s')$  triplets and learns an inverse dynamics model that predicts the action  $a$  that causes a transition  $(s, s')$ . In the post-demonstration phase, given state-only expert trajectories, the model infers the missing actions and fits a BC policy. With this approach they achieved less environment interactions, than methods like GAIL while still achieving comparable performance. Additionally BCO works well across domains and scaled to more complex environments.

Gupta et al. [54] addresses the correspondence problem in IfO. Their method assumes that the expert and agent both have learned a common task beforehand. By using the trajectories from such common tasks, allows to learn a common latent space where behaviour is aligned, while ignoring morphological differences. With that method they were able to successfully transfer skills between torque-driven and tendon-driven arms.

Sermanet et al. [55] proposed a self-supervised solution that deals with viewpoint and morphological differences, called time-contrastive networks (TCN). Given demonstrations from robots and humans, TCNs create embeddings for different viewpoints frames. Frames from the same time point are closer together in the embedding space, while frames of temporal neighbours are pushed apart. Simultaneously TCN learns frame similarities, while also identifying differences between similar-looking frames to create viewpoint invariance. Because the embedding focuses on functional attributes, no explicit human-to-robot joint

mappings required. The reward signal is defined as the distance between agent and human demonstrations.

With the introduction of XIRL (Cross-embodiment Inverse Reinforcement learning), Zakka et al. [56] removed the need for temporal alignment. For that a video encoder is trained on expert demonstrations from many experts, learning an embodiment-invariant embedding that represents the task progress. The task reward is then defined as the distance to the goal embedding, enabling the transfer to new agent morphologies.

Raychaudhuri et al. [52] used an adversarial approach to train a transformation function, that directly transfer the experts behaviour into the agents domain. During inference, they transform each expert transition into the agent’s domain, infer missing actions using an inverse-dynamics model and train the policy with BC. Experiments on six cross-domain MuJoCo tasks demonstrate robustness to dynamics, viewpoint and morphology changes.

Aytar et al. [57] tackle sparse-reward Atari games by aligning YouTube videos with the agent’s observation space. For that they combine temporal distance classification (TDC) for predicting time separation between video frames with cross-modal audio TDC (CMC) for aligning visual and audio information to create a domain-invariant embedding. This embedding captures important game features and events. The agent is then rewarded for reaching a sequence of embedded checkpoints (see figure 4), achieving human-like performance from a single video.

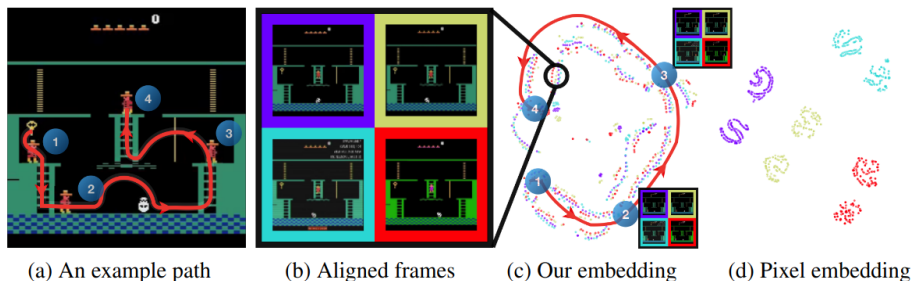


Figure 4: For the path shown in (a), t-SNE projections of trajectories using the embedding space (c) and raw pixels are shown. Four different domains of the same frame are compared side-by-side (b) from the Arcade Learning Environment (purple) and three YouTube videos, showing their trajectories align in the embedded space.

Torabi et al. [58] extend GAIL to state-only data, proposing generative adversarial imitation from observation (GAIfO). In GAIfO the discriminator now distinguishes between state-transitions from the agent and the expert, while the policy learns to fool the discriminator. GAIfO handles both low-dimensional features and high-dimensional vision without time-aligned demonstrations, outperforming BCO and TCN on several benchmarks and performing similar to reward signal trained agents.

Chang et al. [59] formulate IfO as minimizing the Wasserstein distance between state-transition distributions from expert and agent. With their method called observational off-policy Sinkhorn (OOPS), they use an entropy-regularized Sinkhorn distance to obtain a differentiable reward function. Both the policy and transition-embedding network are then optimized with an off-policy RL algorithm. With OOPS the algorithm is scalable. In their experiments, they reached near=expert performance, even with single expert demonstrations.

## 2.5 Diffusion Policies

Diffusion models were first introduced by Sohl-Dickstein et al. [60] as a latent-variable Markov chain motivated by non-equilibrium thermodynamics. The forward (diffusion) process gradually corrupts data  $x_0$  with Gaussian noise

$$q(x_{1:T}|x_0) = \prod_{t=1}^T q(x_t|x_{t-1}), \quad q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I),$$

while the reverse process learns to invert it,

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t), \quad p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)),$$

With improvements like denoising diffusion probabilistic models (DDPM) by Ho et al. [61], diffusion models gained attention. DDPM showed that a model can be trained to reverse the gradual noising process (see figure 5) to generate high-quality samples from high-dimensional data distributions. In parallel [62] developed score-matching, using learned score functions to produce comparable generative results without explicitly learning likelihoods (see figure 5). These diffusion models (also called score-based models) offer stable training, strong mode coverage, providing multi-modality, avoid mode collapse and adversarial training issues [63]. Building upon that Nichol and Dhariwal et al. [64] introduced faster sampling and conditional diffusion techniques that further improved the quality and controllability of diffusion models in image synthesis.

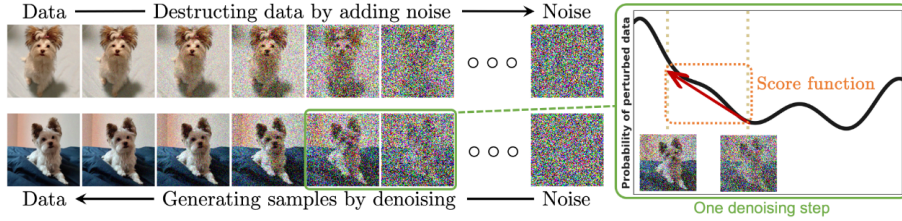
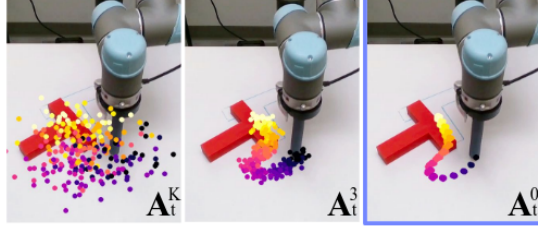


Figure 5: Example of diffusion models perturbing data by adding noise, then reverse this process to generate new data from noise (left). A score function is used for each denoising step in the reverse process (right)



**Output: Action Sequence**

Figure 6: Starting from Gaussian noise, the noise is gradually subtracted, yielding a denoised action sequence

Janner et al. [8] were the first to apply diffusion to decision making. Given an offline dataset of optimal and sub-optimal trajectories, they train a diffusion model over the entire state-action sequences. At inference, reverse diffusion generates candidate trajectories adhering to environment dynamics, while a learned reward gradient steers the denoising toward trajectories that yield higher return. Demonstrating their method on MuJoCo’s HalfCheetah and Walker 2D benchmarks, they were able to synthesize different gaits while finding high-return solutions.

Building on these ideas, Chi et al. [10] introduced an end-to-end, closed-loop robot policy represented as a conditional denoising process, naming it ”diffusion policy”. At each control step they sample an initial noisy action  $a_K$  and iteratively denoise it (see figure 6)

$$a_{k-1} = \alpha_k(a_k - \gamma_k \varepsilon_\theta(o_t, a_k, k)) + \sigma_k z, \quad z \sim \mathcal{N}(0, I),$$

where

- $\varepsilon_\theta$  is a U-Net or transformer that predicts the noise conditioned on the current observation  $o_t$

During training, the noise predictor treats each partially corrupted action  $a_t(k)$  as input and learns, via noise-conditioned score matching to predict the expert action that would have produced that sample, efficiently performing behaviour cloning at every noise level. At deployment, a receding-horizon variant of the reverse diffusion process generates short action sequences, allowing for real time planning. This approach consistently outperformed previous IL methods on twelve simulated and physical manipulation tasks. Because of the generative formulation of diffusion policies, they are also able to model multi-modal action choices without collapsing into a single mode (see figure 7).

Concurrently [65] demonstrated that the same noise-prediction approach as in [10] also captures the stochasticity of human demonstration data in games and simulation better than BC or adversarial methods.

Despite their promising results in policy generation, diffusion policies still face practical issues. The iterative denoising process (usually 20-100 steps)

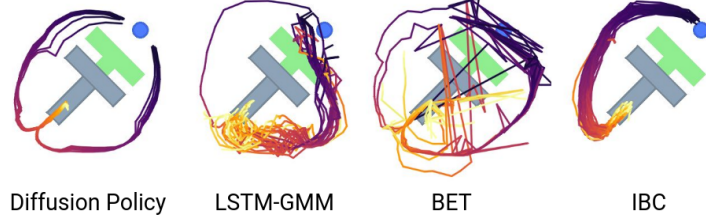


Figure 7: At the given state, the end-effector (blue) can either go left or right to push the block inside the goal state. Diffusion policy learns both trajectories and commits to one during roll-out, while other methods fail to create these consistently.

is slower than single feed forward methods. Although recent solvers such as DDIM [66] or DPM-Solver [67] decrease the number of steps to a single digit number, they do not eliminate them. In domains like robotics, that require sub-millisecond control, running multiple diffusion steps is infeasible, even with acceleration through solvers. Since diffusion models learn an entire action distribution, they need sufficiently large datasets. With limited or unbalanced data, they may either fail to learn policies or collapse to more frequent ones. Due to the black-box nature of these models, application in safety-critical environments is infeasible, since there is no formal guarantee, that sampled actions satisfy hard constraints, although the integration of control barrier functions [68] shows promising results (see figure 8)

## 2.6 Behavioural Transformers

After the landmark paper "Attention is all you need" by Vaswani et al. [69] was published, transformer architectures gained popularity, even beyond scientific research. By now, transformers are widely present in everyday tools such as OpenAI's ChatGPT models [70] or Anthropic's Claude series [71].

A transformer is a neural network architecture, that unlike recurrent neural networks (RNNs) and convolutional neural networks (CNNs), processes an entire sequence of tokens in parallel. Given a sequence of  $T$  token embeddings

$$X = (x_1, \dots, x_T) \in \mathbb{R}^{T \times d},$$

the model first projects  $X$  to queries  $Q$ , keys  $K$  and values  $V$ . To measure how relevant one token is to another, it computes pairwise attention scores

$$score(i, j) = \frac{Q_i K_j^\top}{\sqrt{d_k}},$$

and turns them into weights with a softmax

$$\alpha_{ij} = \text{softmax}_j \left( \frac{Q_i K_j^\top}{\sqrt{d_k}} \right).$$

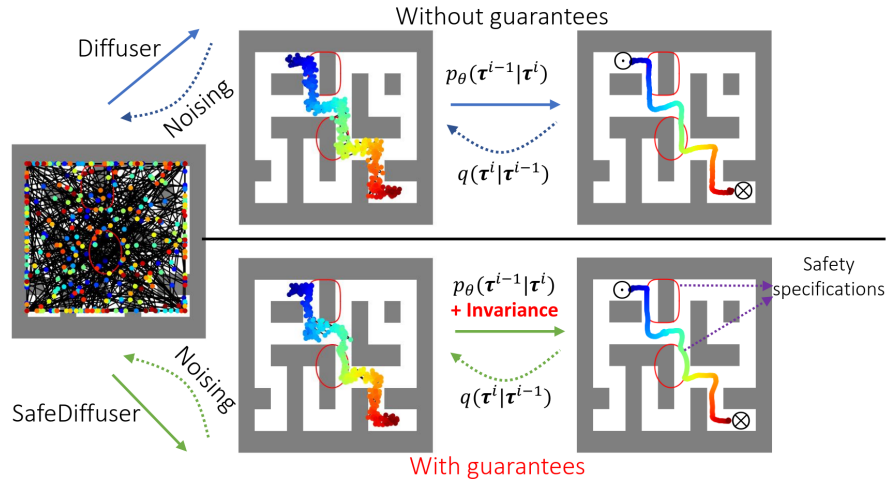


Figure 8: Standard diffusion planning (Diffuser) (top) intrudes safety areas (red circles), while SafeDiffuser (bottom) adheres to safety specifications, avoiding to enter safety areas

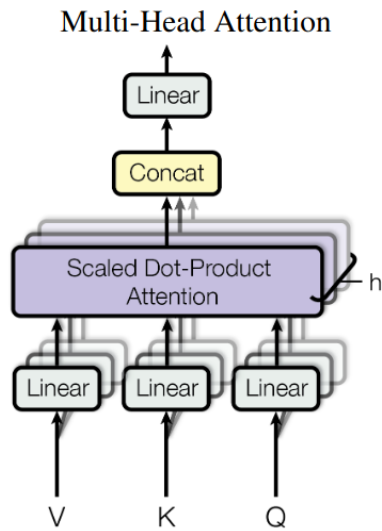


Figure 9: Multi-Head Attention consists of several attention layers running in parallel

Each output vector is then a weighted sum,  $z_i = \sum_{j=1}^T \alpha_{ij} V_j$ , so information can flow between nearby and distant positions without an sequential bottleneck. The core of transformers are the multi-head self-attention blocks.  $h$  separate heads repeat the calculation above in parallel, producing  $h$  different  $\{z_i^{(h)}\}$  and concatenate them, yielding many views of the same sequence (see figure 9). Because the whole sequence is visible at once, the model can figure out how different tokens relate to each other and therefore excels on tasks that need a lot of context [72]. During inference, the transformer predicts a probability distribution over the next token,

$$P_\theta(x_{t+1} \mid x_{1:t}),$$

selects this token and appends it to the context. In tasks like text generation [73] this token generation repeats an auto-regressive loop to create predictions as continuation of the input text. This parallel processing makes transformers highly scalable and a key technology in many areas like speech and language processing [74].

Behavioural Transformers (BeT) were introduced by Shafiullah et al [9] adopting the idea of token prediction to offline IL tasks by using transformers to predict action sequences, that mimic expert demonstrations. They formalized this as maximum likelihood behaviour cloning (MLBC), where the goal is to find a policy that maximizes the probability of observed state-action pairs from a dataset

$$\theta^* = \arg \max_{\theta} \sum_{(s_{1:T}, a_{1:T}) \in \mathcal{D}} \sum_{t=1}^T \log \pi_\theta(a_t \mid s_{1:t}, a_{1:t-1}).$$

Unlike classical BC, the predicted action depends on a history of observations  $h_t = (s_{t-h:t}, a_{t-h:t-1})$ , making it non-Markovian. To handle continuous spaces and capture multi modality, BeT clusters actions into  $k$  bins (with  $k$ -means),  $c(a) \in \{1, \dots, k\}$ , predicts a categorical distribution  $p(c|h_t)$  and learns a small residual  $\delta \in \mathcal{R}^{dim A}$  to compensate for the small error from the cluster center such that  $\hat{a}_t = A_c + \delta(h_t, c)$ . Because BeT uses cross-entropy loss, the model is rewarded for covering all modes in the demonstrations, which avoids the do-nothing averaging behaviour of typical mean-squared-error BC (see figure 10).

Wu et al. [75] observed that real datasets usually contain both optimal and sub-optimal trajectories. Introducing Similarity-Weighted Behaviour Transformers (SWBT) they train a transformer on all demonstrations, then compute a quality score and re-weight the loss from BeTs accordingly. This puts more weight to higher-quality example making them influencing the final policy more. On the ManiSkill2 benchmark [76] SWBT achieved significantly higher success rates across all tasks, compared to methods like BC and BeT.

Weaver et al. [77] proposed behavior transformer adversarial imitation learning (BeTAIL). BeTAIL combines BTs sequence modeling with AIL to create racing policies from human demonstrations. In a two step approach, they first train an offline BeT as an expert prior and an online AIL-based residual policy,

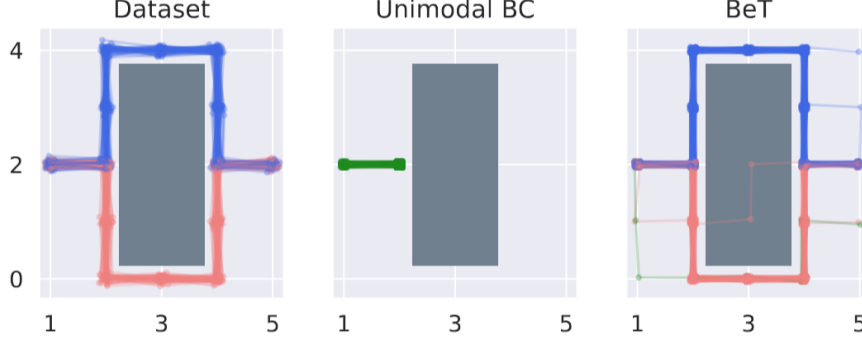


Figure 10: While regular MSE BC (mid) takes 0 actions to minimize MSE, BeT (right) captures multi-modal distributions

such that the expected action is  $a_t = a_t^{BeT} + a_t^{res}$ . The residual policy corrects distribution-shift errors that occurs during offline learning, leading to smoother steering, faster lap time and better adaptivity to out-of-distribution states, in Gran Turismo Sport, compared to pure AIL approaches.

Nevertheless, transformer architectures come with some challenges. Transformer models are very data-hungry and training BeTs often requires a large amount of data [75]. If only a few demonstrations are available BeTs tend to overfit or fail to generalize. Furthermore, since BeTs are trained by pure offline imitation, there is no guarantee of performance outside the distribution of its training data, making them also suffer from covariate shift [77]. Additionally, they also require a high computational and memory cost, because of their quadratic scaling of the input sequence length. Also similarly to diffusion policies, transformers also lack interpretability due to their black-box nature.

### 3 Applications in Robotics

After giving a broad overview of a variety of imitation learning methods, we will discuss robotics application in the following part of this review. We discuss a variety of recent robotic advancements in different fields, utilizing imitation learning approaches.

#### 3.1 Robot Manipulation

Robot manipulation enables robotic systems to physically interact with objects and their environment to perform precise, coordinated tasks. Effective manipulation requires addressing challenges such as accurately perceiving objects, robustly grasping items of various shapes and sizes, coordinating multiple limbs and adapting to dynamic and uncertain environments.



With hierarchical deep relational imitation learning (HDR-IL), Xie et al. [78] introduced a method for two-arm robots to learn from demonstration data only, while still generalize to unseen objects and robot configurations. Their HDR-IL consists of two levels of planners that are trained with BC. The first level is a high-level planner that selects the next motion primitive (e.g. grasp, move, lift, etc.), given the recent state history. The second level is a primitive-specific dynamic module, which uses an inverse kinematic solver, turning the selected motion primitives into joint commands. The datasets were generated by the PyBullet simulator with a Baxter robot. Their findings were, that HDR-IL compared better, than any other IL baseline in their work.

Liu et al. [79] address the challenge data-scarcity, multi-modal action complexity and noisy robotic signal in bimanual robots. They introduce robotic diffusion transformer (RDT-1B), a diffusion-based policy that predicts action chunks (64 step sequences) using a with a diffusion transformer feature-extractor with robot-specific settings. At test time RDT-1B samples each chunk at 6Hz. They pre-trained their model on more than 1 million single-arm trajectories from 64 diverse datasets and then fine-tune on  $\sim$  from 46 single-arm datasets and fine-tune on  $\sim$  6000 high-quality dual-arm demos. On real-robot manipulator benchmarks, RDT-1B delivers an increase in success rate over the strongest previous methods (ACT, OpenVLA, Octo), achieve zero-shot generalization. Additionally they are also able to master further complex tasks by fine-tuning with just 1-5 demonstrations.

Haldar et al. [80] tackled the few-shot imitation learning gap by teaching robots robust vision-based manipulation skills in under a minute of demonstration per task. Their method fast imitation of skills from humans (FISH) first trains a lightweight CNN encoder and a non-parametric base policy directly on the short demonstration frames with BC. During online interaction, their parametric residual policy (an actor-critic with n-step DDPG) adds an offset to the action, outputted by the CNN. They tested FISH on three different robot platforms, each on three different tasks (see. figure 11, demonstrating that FISH was able to solve every task with less than 1 minute of demonstration, while also generalizing to unseen situations.

### 3.2 Mobile Robot Navigation and Autonomous Driving

Mobile robot investigates ground and aerial vehicles to operate and move partially or fully without human intervention. Therefore it integrates techniques from robotics, artificial intelligence and control theory to provide self-driving capabilities, usually in wheeled robots. Especially in autonomous driving, safety plays a critical role to ensure accident free transports, but also delivery robots face the challenge of safely interacting with their surroundings.

The idea of autonomous driving already dates back to 1988, where Pomerleau et al. [3] proposed ALVINN, the first vision-based system to map images directly to steering commands, utilizing a shallow three-layer NN to perform road following.

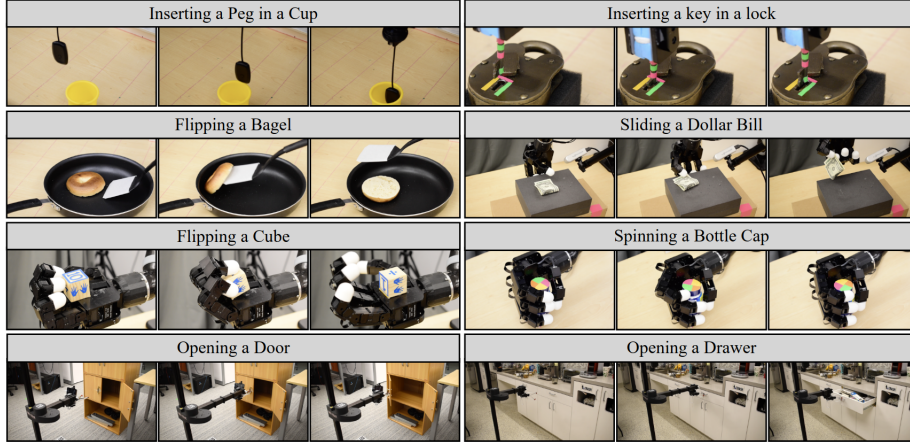


Fig. 5: A visualization of rollouts from FISH on a selected set of 8 tasks.

Figure 11: Visualization of roll-outs from FISH on a selected set of 8 tasks

Bojarski et al [13] propose one of the first end-to-end steering model for self-driving cars. They tackle the existing problem of self-driving systems, relying on modular pipelines that break tasks into individual problems, by replacing it with a convolutional network that predicts steering directly from RGB frames. For training, they collected 72 hours diverse driving data, augmenting them additionally with synthetic shifts, rotations and steering perturbations to promote agent recovery. The CNN is trained with BC. Before real-world deployment, performance is evaluated in simulation by using pre-recorded video streams. The performance is quantified as 'autonomy measure', describing the vehicles ability to drive without human intervention. Assuming that each human intervention averages to 6 seconds, they defined the autonomy measure as

$$autonomy = (1 - \frac{number\ of\ interventions \cdot 6seconds}{elapsedtime[seconds]}) \cdot 100.$$

In real-world tests, the network demonstrated high-performance with approximately 98% autonomous driving.

Pan et al. [81] trained an end-to-end imitation learning policy for off-road autonomous driving, that solely relies on low-cost onboard sensors. To train the policy, they used two different imitation learning approaches, DAGger and a batched version of BC. The expert data is collected by running demonstrations, using a model predictive controller (MPC). Although the agent eventually operates without expensive sensors, the MPC expert uses a full sensor suite (including GPS and (inertial measurement units (IMUs)) to make accurate state estimation. They deployed their network on a 1/5 scale AutoRally car, equipped with camera and wheel speed sensors on a elliptical dirt track. The agent policy was able to operate at a frequency of 50Hz. Performance was measured by metrics like average speed, top speed and completion ratio. In their results, they

compare both of their IL approaches, showing that the online DAgger approach yields closer performance to the expert, demonstrating a 100% completion ratio in the experiments, while BC reaches similar speed metrics, but crashes in less explored environments.

Karnan et al. [82] introduced visual-observation-only imitation learning for autonomous driving (VOILA). Challenging autonomous navigation from video-only observation data. This is particularly important, when demonstration videos are recorded with different devices, outside of the agent’s onboard cameras, leading to a viewpoint mismatch. VOILA uses a reward function, based on keypoint matching. Keypoints are similar points of interest in video data, that might be shared between videos, to create an alignment of viewpoints. For keypoint extraction, they used SUPERPOINT [83], known for robustness in perspective, scale and lighting. The reward function is then defined as

$$R = \begin{cases} F + V - \lambda||a||, & \text{alive} \\ -10, & \text{done} \end{cases}$$

where

- $F$  is the reward defined as the ratio of keypoint matches between agent observations and expert demonstrations
- $V$  reward for performing similar observation transitions than the expert
- $a$  action penalty for making large steering changes

If the robot is in the done state, it has either crashed or the number of keypoint matches, dropped below 10, otherwise the agent is in the alive state. They authors conducted experiments in the AirSim simulator [84] on multiple indoor and outdoor road segments. Using the Hausdorff distance to measures performance between expert and agent trajectory, their findings were that VOILA successfully imitated the expert demonstration and generalized well to unseen tracks. In circumstances, where the expert and agent viewpoints differed, VOILA maintains good performance, while GAIfo failed to imitate effectively. The method is further validated on a real robot, performing a hallway patrol task in an indoor office environment. Here demonstrations are recorded from a handheld mobile phone and compared to demonstrations collected with the onboard cameras (see 12). VOILA is able to learn an efficient imitation policy, even when trained with viewpoint mismatches data, while handling perturbations. The method is also extends further to learn navigation policies by using LIDAR data, leading to faster convergence with vision-based inputs.

Wang et al. [85] addressed the challenge of designing decision-making systems for autonomous driving in busy environments for lane-changing maneuvers. For that they introduce an adversarial method, called augmented adversarial inverse reinforcement learning (AugAIRL). This method builds on AIRL to learn both policy and underlying reward function. They augment AIRL by incorporating semantic, domain-specific reward terms for successful lane switching,

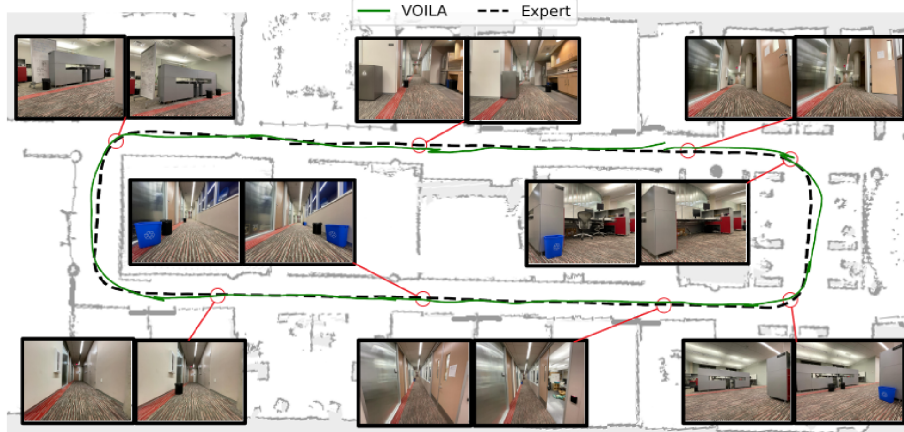


Figure 12: Agent deployed in real-world environment. The learned policy is robust to visual differences between training and deployment environment. In each picture the left image shows the training environment and the right image the perturbed environment

expert imitation, collision, safety margins, proper lateral movement and for carefully interacting with other vehicles. They incorporate these rewards into the discriminator and the agents network to ensure that both are constantly influenced by the semantic feedback. Their findings are that AugAIRL outperforms other imitation learning methods like AIRL and GAIL. The augmented method demonstrates faster convergence and yields more stable training results with closely matching the expert performance.

To study human-aware robot navigation in busy environments while respecting social norms and ensure safe interaction, Karnan et al. [86] released the Socially Compliant navigation Dataset (SCAND) dataset, because of the lack of large-scale real-world social navigation data. The authors identified, that there are multiple strategies to navigate in a socially compliant manner, depending on the context. Recognizing these strategies is crucial for developing more flexible and robust navigation policies. Their data collection platform consists of a wheeled Clearpath Jackal and a legged Boston Dynamics Spot, using multiple sensors like 3D LiDAR, Cameras, IMUs, Odometry and Joystick command recordings. The dataset contains 8.7 hours of recordings, 138 trajectory and covers 25 miles of navigation in both indoor and outdoor settings. Each trajectory is labeled with one or more of 12 predefined social interactions to capture the social context (see table 3.2). A BC policy that jointly predicts global (socially compliant) path and local velocity commands was compared the MoveBase [87] baseline, reaching significant reduction of the Hausdorff measure. On real-world tests with human participants, the robots are also rated more socially compliant (4.39/5 points) and safer (4.71/5 points).

Zheng et al. [11] introduce Diffusion Planner, a motion-planning framework

| Tag                             | Description                                                                     |
|---------------------------------|---------------------------------------------------------------------------------|
| Against Traffic                 | Navigating against oncoming traffic                                             |
| With Traffic                    | Navigating with oncoming traffic                                                |
| Street Crossing                 | Crossing across a street                                                        |
| Overtaking                      | Overtaking a person or groups of people                                         |
| Sidewalk                        | Navigating on a sidewalk                                                        |
| Passing Conversational Groups   | Navigating past a group of 2 or more people that are talking amongst themselves |
| Blind Corner                    | Navigating past a corner where the robot cannot see the other side              |
| Narrow Doorway                  | Navigating through a doorway where the robot waits for a human to open the door |
| Crossing Stationary Queue       | Walking across a line of people                                                 |
| Stairs                          | Walking up and/or down stairs                                                   |
| Vehicle Interaction             | Navigating around a vehicle                                                     |
| Navigating Through Large Crowds | Navigating among large unstructured crowds                                      |

Table 1: Descriptions of labeled tags contained in SCAND

capturing the multi-modal nature of human driving, avoids rule-based post-processing that most learners still require for ensuring safety and allowing on-the-fly customization of driving styles (e.g. safer, smoother, faster) without re-training. They combine both prediction and planning into a single task by generating a set of future trajectories of the agent and the  $M$  closest cars. Their model, which is based on diffusion transformers [88]) uses a noised future position of the agent and the neighbouring cars, 2 seconds of observation history, vectorized lanes and traffic lights and a coarse navigation route as input. They use an MLP for encoding the history and the map information to fuse them into ‘scene tokens’. These tokens are combined with the vehicle tokens (tokens, that encode the state of the car and  $M$  closest neighbours) via cross-attention. During training, they use a denoising loss for diffusion models. During testing, they speed up the sampling rate with DPM-Solver++, to 10 diffusion steps, yielding a 9 second plan in about 0.04 seconds. During inference, they can steer

the behaviour of the trajectory generation by adding simple cost terms (e.g. stay further away from others, keep inside drivable area, aim for 12 m/s, etc), without the need of retraining, running at 20Hz on a single GPU. Comparing diffusion planner on the nuPlan simulator [89] (Val14, Test14, Test14-hard) and on a fresh 200-hour delivery robot dataset revealed that diffusion planner scored best among previous learning models, while matching rule systems. Adding a trajectory-offset refiner increases the score on the nuPlan dataset even more. On the delivery dataset, they outperformed other methods as well.

### 3.3 Legged Robotics

The field of legged robotics focuses on design and control of robots using articulated limbs. In contrast to wheeled or tracked mobile robotics, legged systems can walk, run or hop across uneven, discontinuous and structured terrains.

With DeepMimic by Peng et al [90], a first large-scale framework for end-to-end learning of naturalistic motion in simulation. During each training episode, a random motion-capture clip is sampled uniformly from a set that covers more than 25 skills. The agents were rewarded for matching joint orientations, velocities, end-effector positions and center-of-mass from the reference motions. By training separate, independent policies for all of the different skills, a composite controller selects between them using a soft-max function, producing smooth transitions. After successfully learning over 25 skills, they incorporated complex tasks into their framework like direction heading, ball throwing or terrain traversal across a variety of obstacles. For that they used a combined reward function

$$r = (1 - \alpha)r_I + \alpha r_T$$

where

- $r_I$  is the imitation reward
- $r_T$  is the task reward
- $\alpha$  is a tradeoff parameter (they used  $\alpha = 0.5$ )

By training these individual skills beforehand, they were now able to solve these more complex tasks successfully, without retraining any of the skills.

The same authors later extended this approach to Laikago hardware. Based on animal motion capture data, Peng et al. [12] converted animal motion data to fit the robot’s morphology using inverse kinematics. For training they incorporated strong randomization of masses, inertias and contact parameters and trained an information-bottleneck encoder, that embeds those dynamics into a latent embedding. During deployment around 50 trials of advantage weighted regression were needed to search the latent space to recover agile gaits. They successfully trained the Laikago robot to perform various gaits and dynamic behaviours like turning and hopping in simulation and on real hardware. This hierarchical control (see figure 14) outperformed robust baselines in sim-to-real transfer.

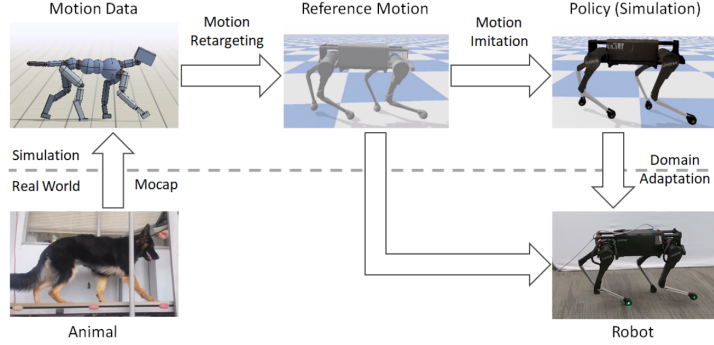


Figure 13: The three stages of motion targeting, motion imitation and domain adaptation are visualized here. The robot receives a recorded input motion data from animals and outputs a control policy in real robots

With the adversarial motion priors (AMP) approach, Peng et al. [91] addresses the fundamental challenges of creating natural life-like behaviour and limitations of traditional tracking-based methods. Using GAIL, they imitate skills from motion data, their so called motion priors, without the need of precise temporal alignment or synchronization with reference motions or specific reward functions. To transfer learned skills to more complex tasks, they combined the task reward with the AMP reward with a weighting

$$r = (1 - \alpha)r_{AMP} + \alpha r_{task}, \quad \alpha \in [0, 1]$$

to train their tasks (see. This AMP approach demonstrates capabilities in motion generation, skill composition and transfer, similar to [90] and [12], while producing more realistic motions.

Li et al. [92] introduced a method for imitating agile robotic skills from rough, partial and potentially physically incompatible demonstrations. For that they introduced Wasserstein Adversarial Behaviour Imitation (WASABI), which adapted GAIL approach by replacing the Jensen-Shannon divergence with the Wasserstein distance. They evaluated their method on a Solo 8 quadrupedal robot [93], teaching various dynamic skills like leaping, wave-like locomotion and performing backflips. WASABI outperformed the least-squared GAN (LSGAN) and the traditional GAN approach. Notably, WASABI achieved successful sim-to-real transfer with policies trained in simulation and enabled zero-shot skill transfer from Solo 8 to the ANYmal [94] platform.

Yang et al. [95] introduced versatile instructable motion priors (VIM), an adversarial framework enabling the combination of animal motion capture data and diffusion-generated synthetic clips inside a single controller. The VIM architecture consists of a motion encoder, mapping motion segments into a latent skill space with an information bottleneck for maintaining temporal consistency and a dual reward, describing functional (foot placement, body height, etc.) and

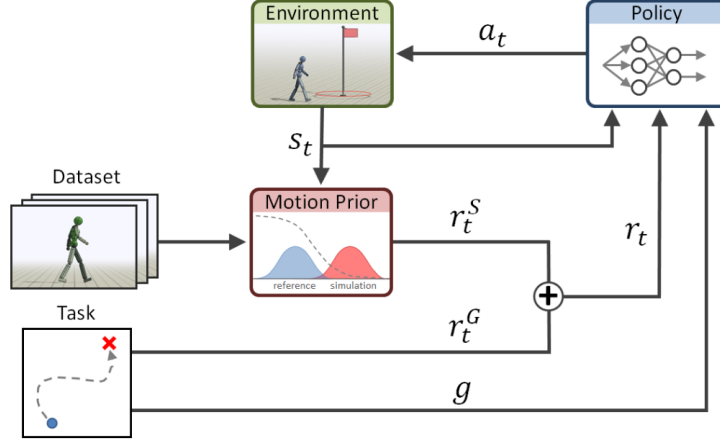


Figure 14: Using a dataset to learn motion priors beforehand, a policy can be trained that interacts with the environment, using motion priors, to learn complex tasks

stylistic (adversarial discriminator feedback and joint-angle tracking) properties. VIM outperformed baselines like GAIL and WASABI, while also demonstrating real-world transfer of skills.

Vogel et al. [96] showed the first quadruped that can walk normally and climb industrial ladders with 90% zero-shot success. Therefore each foot was fitted with a lightweight hook (see figure 16) and a two-stage teacher-student pipeline was trained entirely in simulation. The teacher policy is trained with RL in a privileged, noise-free setting, receiving perfect sensor data, access to a local height scan, exact contacts, external forces and the true ladder geometry. Because the task is now a fully observed MDP, the teacher quickly discovers efficient motions that combine hook-assisted climbing. The student policy only sees what the noisy hardware will provide, turning the problem into a noisy POMDP. The student imitates the teachers joint commands via BC and reconstructs the teachers hidden state, using a trained "belief encoder". Adding further heavy domain randomization of masses, friction, sensor noise, external pushes and ladder parameters during student roll-outs, forces the network to generalize. After several days of simulating, the resulting policy is transferred to hardware (see figure 15), successfully allowing the robot to climb the ladders 232 times faster, than previous approaches, with a 90% success rate on a zero-shot manner.

### 3.4 Aerial Robotics

Imitation learning for aerial robots deals with the inherently underactuated and nonlinear dynamics of flying platforms making stable policy generation challenging [97]. Vision based demonstrations further increase the difficulty and given





Figure 15: ANYmal robot in 3 different states during ladder climbing, showcasing the movement flow.

their high-dimensional image streams under latency and bandwidth limitations, decreases sample efficiency and prevents real-time development [98]. Moreover, bridging the sim-to-real gap by learning from imperfect aerodynamic modeling, sensor noise and environmental disturbances, yields strong distribution shifts, that decrease performance when transferring policies.

High speed quadrotor racing demonstrates the challenge of end-to-end visuomotor imitation learning, as the platform must process high-dimensional RGB observations while producing fast control output. Xing et al. [98] combine the sample-efficiency of IL with the asymptotic performance of RL to address this challenge. In their framework. A teacher policy is trained in simulation using the full state and a dense reward. A student policy then receives either an embedding, of the last 4-64 RGB frames or gate-corner pixels (a gate in drone racing is a rectangular hoop, that the drone must fly through). Using DAGger, the student policy is then trained to mimic the teacher policy. Finally, the student policy is fine-tuned RGB frames only. They evaluated their policy on three different simulated tracks ("Split-S", "Figure-8" and "Kidney") (see figure 17) for 10 laps or 50 seconds. Their method consistently surpassed methods like BC, pure RL, DAGger and PIRLNav [99] baselines in simulation. Real world flights on the same tracks showed the transfer ability and improved lap times, compared to DAGger (see figure 18).

Wang et al. [100] tackle autonomous UAV navigation along cluttered rivers using only onboard vision. In their initial release, the authors made an publicly

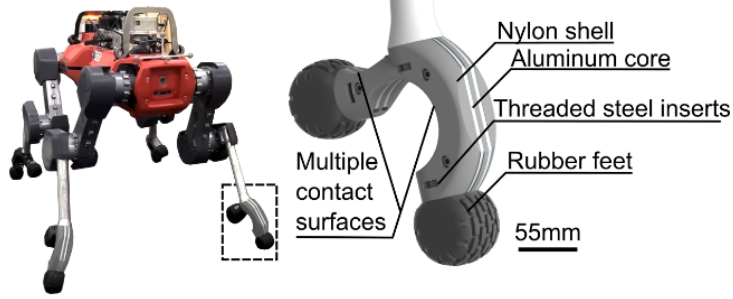


Figure 16: (left) ANYmal robot (right) hook end-effector to enable locomotion as well as grappling to ladders

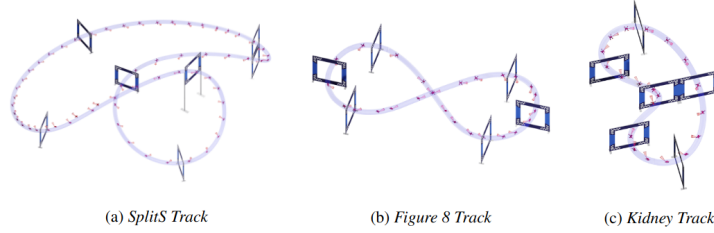


Figure 17: Visualization of the drone racing tracks used for experiments

available photo-realistic Unity environment that models bridges, variable river widths with side rivers and 3D flight dynamics. The challenge is described as a non-Markovian track-following problem, where the UAV earns a reward only for navigating into previously unseen river segments and is penalized for collision, yaw deviation or stagnation. With  $n$  being newly visited river segments and  $N$  being the total number of river splines, the reward is defined as

$$r(s, a) = \begin{cases} -1, & \text{if failure} \\ 10 \frac{n}{N}, & \text{if new visited segment} \\ 0, & \text{otherwise} \end{cases}$$

To overcome sample-inefficiency and the limited state coverage of BC, the authors proposed a synergistic RL + IL framework. Here a policy is pre-trained on 50 human recorded trajectories using BC. During training a PPO agent fine-tunes the policy and whenever an episode is completed, with a reward  $r > 4$ , the trajectory gets appended to the expert demonstrations. They called this aggregation of new trajectories in to the demonstration buffer dynamic BC. In their experiments, they compare PPO, StaticBC, DynamicBC, PPO+ StaticBC and PPO + DyanmicBC on the Unity river environment and on a partially observable CliffCircular grid world. PPO+DynamicBC converges fastest and achieves the highest final reward in the river as well as in the CliffCircular world.

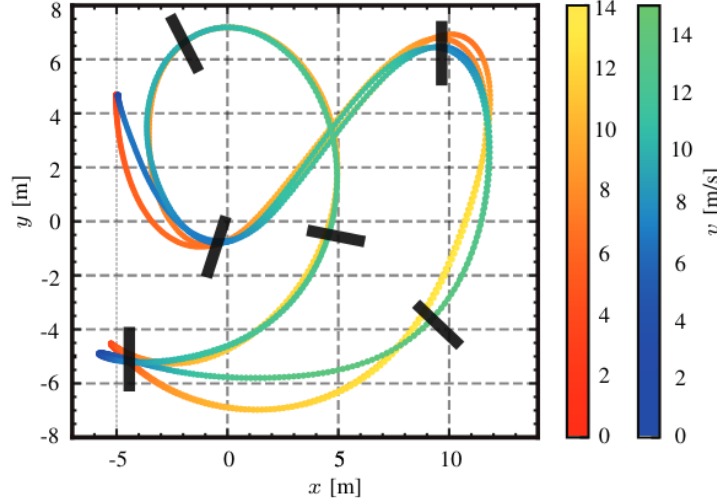


Figure 18: Real world comparison of trajectories between DAgger (orange) with an avg. lap time of 8.06s and bootstrap RLIL (green) with an avg. of 7.67s lap time



Figure 19: Visual input of drone agent as concatenation of six camera images

### 3.5 Swarm Robotics

Swarm robotics studies how large numbers of relatively simple robots can collectively perform tasks that would be too difficult or even impossible for a single robot, by drawing inspiration from biological systems, such as ants colonies or fishes. Since such behaviours originate from simple local interactions among agents, even propagating through the whole network makes it difficult to replicate such behaviours. Recent advances are leveraging imitation learning to automatically infer coordination policies from expert demonstrations.

Schilling et al. [101] present the first end-to-end vision-based control policy for fully decentralized quadrotor swarms. Instead of sharing GPS or motion-capture data, every drone predicts its own 3D velocity purely from six low-resolution omnidirectional grayscale images (see figure 19). For the expert behaviour they used the classic Reynolds flocking rule [102], with an additional migration term that adds a goal-directed attraction force to each individual agent. A CNN is trained with DAgger to mimic those expert velocities and when deploying on real-world hardware, additional DAgger iterations are exe-

cuted. In simulation with nine drones, the agents show that the flock integrity is preserved. When deploying the policy on the 2 real world agents, the drones successfully show the desired circle (see figure 20, carousel and push-pull (see figure 21 flight trajectories.

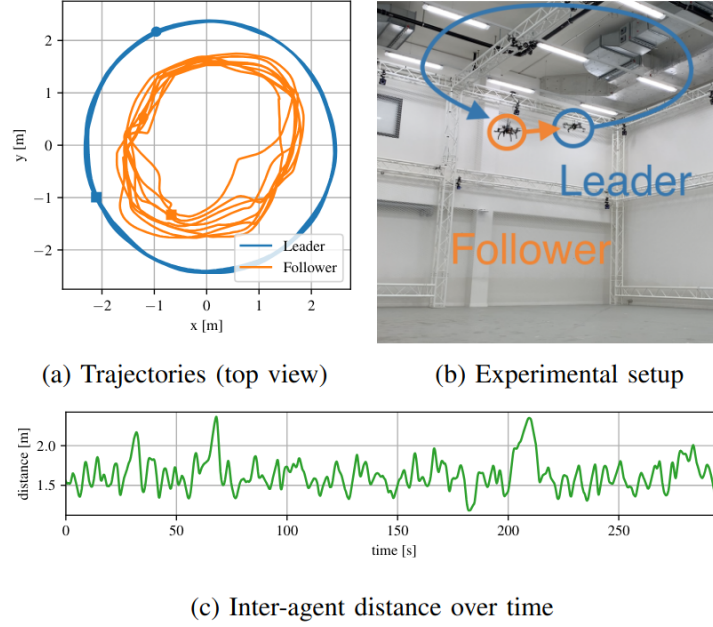


Figure 20: In the circle experiment (b), the follower drone has to keep constant distance from the leader drone, following the leader in a circular trajectory. (a) shows the flight trajectories of leader and follower horizontal xy-plane. (c) shows the followers distance to the leader.

Further contributions to swarm robotics were made by Wu et al. [103], who introduced an AIL framework by plugging a modular deep-attention network into GAIL, making policies scalable to different swarm sizes and interpretable. They validate that AIL plus attention returns sample-efficient and explainable swarm controllers.

Agunloye et al. [104] tackle the problem of reconstructing swarm behaviours without looking at the original controllers by transforming demonstration trajectories into explainable cohesion-based features (distance to all entities) and matching them via multi-agent GAIL. They show that learning of obstacle avoidance, homing and mimicking the demonstrations is possible in a shared environment

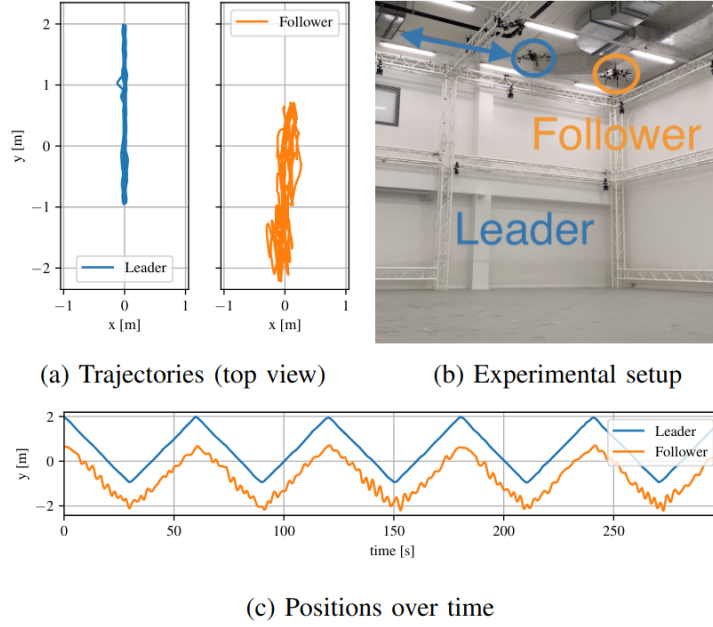


Figure 21: In the push-pull experiment (b), the leader repeatedly flies towards and away from the follower, while the follower needs to maintain constant distance. (a) shows the drones trajectories in the horizontal xy-plane, while (c) shows the the y-position over time, indicating successful task graduation

### 3.6 Agricultural Robotics

With the ever-growing need for agricultural supplies [105], farms are turning to robotic systems to boost productivity while using fewer resources. Agricultural robots range from autonomous tractors and spraying drones to fruit picking quadrupeds, combining computer vision, advanced sensors and machine learning approaches to perform tasks faster and more accurately than humans alone. By automating these jobs, they could help to chemical inputs and capture detailed field data in real time. Ultimately, agricultural robotics is emerging as a key for sustainable farming that can keep pace with rising global food demands.

Industrial mushroom farms still rely on human pickers. Because of the delicate tissue harvesting requires a careful sequence of hand motions. To solve that Porichis et al. [106] introduced an end-to-end visual IL pipeline, that enables a robot to perform the complex task of harvesting mushrooms directly from camera input. To avoid online exploration, the authors split the policy into two modules, trained entirely offline. The first module, serves as a representation learner, using a convolutional variational autoencoder (CVAE) [107] to embed the pictures from the gripper camera into a latent space. The second module is trained by BC, mapping the embeddings directly to an end-effector action.

Expert data were generated in simulation with a rule-based controller, picking mushrooms. With their method, they achieved 96% success rate on 50 roll-out in simulation. Tests on hardware, were not conducted.

Kim et al. [108] tackled the problem of pepper harvesting in open field conditions. Using diffusion policies they solved the problem of robotic harvesting in open field conditions (see figure 22a). This is particularly difficult, since environmental changes like illumination, wind, highly variable fruit posture and foliage, obstructing visual sensors, make reliable fruit detection and picking complicated. In their work, they collected 300 real-world demonstrations with a custom handheld gripper (see figure 22b) and trained a diffusion policy to map sequences of fisheye images to future joint commands. The learned policy was deployed on a UR5e-arm mounted on a Clearpath Husky UGV (see figure 22a, c) Across 221 field trials, the robot achieved a success rate of 28.95% for approaching, cutting and grasping the pepper, without human intervention on the raw field.

As Kim et al. (2024) have previously shown, occlusion can significantly impede the efficiency of fruit-picking. Li et al. [109] tackle this problem of occlusion in fruits and vegetables harvesting by learning an active-vision policy that repositions a camera, mounted on a manipulator to find better viewpoints, preferably with less occlusion. For that they collect 50 human demonstrations (in simulation) representing the trajectories to the optimal viewpoint. With the collected data, they trained an action chunking transformer (ACT) BC policy. ACT predicts a chunk of 5 actions ahead, giving smooth, precise motion. A conditional VAE with a ResNet-18 image backbone and transformer encoder/decoder minimizes

$$\mathcal{L} = \mathcal{L}_{recon} + \beta \mathcal{L}_{KL}, \quad \beta = 10,$$

where

- $\mathcal{L}_{recon}$  is the reconstruction loss, predicting how close the predicted action-chunks are to the expert demonstrated actions,
- $\mathcal{L}_{KL}$  is the KL divergence loss from the conditional VAE,
- $\beta$  a scalar weighting term. Setting  $\beta = 10$  prioritizes a smooth latent space over exact action matching,

and a temporally decayed ensemble, fusing overlapping action chunks (generated by the ACT) at test time. They achieved a control frequency of 10Hz and reached an unobstructed view, within 3.1 seconds on average with an 86.7% success rate on 15 unseen Gazebo scenes. Experiments on real hardware, achieved 66.7% success.

### 3.7 Construction Robotics

Robotics in construction and infrastructure aim to enhance the efficiency, precision and safety in dangerous and demanding environments by using task automation.



Figure 22: (a) Robots face domain-specific challenges, including variable plant morphology, unpredictable lighting and unstructured field conditions. (b) Collecting demonstrations with a custom handheld gripper device (c) Trained policy enables autonomous pepper harvesting

Autonomous excavation brings the advantage of robotic automation, improved safety and round-the-clock productivity to heavy construction and mining sites. Chen et al. [110] proposed excavator action chunking with transformers (ExACT), the first end-to-end controller that maps sensor observations directly to hydraulic valve commands for a 21 ton excavator. With the input of a front-facing RGB image, two LiDAR elevation maps for the digging and dumping zones and joint angles, ExACT outputs a chunk of future valve actions (similar to [109]) with a control frequency of 10 Hz. They tested their model on three different tasks with increasing difficulty, where the ‘reach’ task is defined as reaching a target position, the ‘dig\_dump’ task is defined as digging soil in a certain area and dumping it in another area and the ‘dig\_dump\_return’ task is defined the same as ‘dig\_dump’ with the addition of returning the excavator bucket back to the digging area (all in simulation). On the ‘reach’ task, they consistently aligned with the goal position, on the ‘dig\_dump’ task Ex-



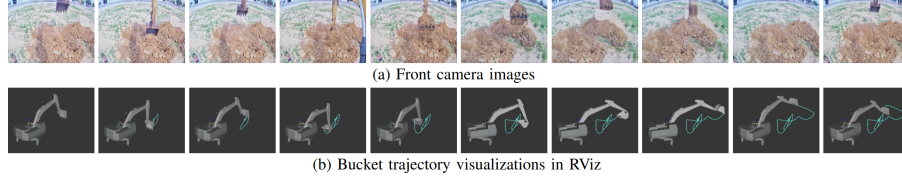


Figure 23: Front camera images (a) and bucket trajectory (b) during testing of the ‘dig\_dump\_return’ task

ACT reproduces the digging cycle, occasionally overshooting the soil release and on the ‘dig\_dump\_return’ task, the precisely completed the task (but with joint-position control, due to the imperfect performance of valve control in the ‘dig\_dump’ task) (see figure ??).

### 3.8 Soft Robotics

Soft robotics is a branch of robotics that focuses on designing and controlling robots made from compliant materials, including, but not limited to silicones, elastomers and fluidic networks. These robots can bend stretch and morph like living organisms. By replacing rigid joints with soft structures, they outperform at gentle grasping, safe human-robot interaction and navigating complex, unpredictable environments that defeat traditional rigid-bodied robots.

Soft continuum manipulators offer advantageous compliance and adaptability, but its virtually infinite degrees-of-freedom makes RL control very sample hungry. Nazeer et al. [111] introduces Soft DAgger, a sample efficient IL framework for soft robots. Their goal was to teach a pneumatic soft-robotic arm to “write” letters and draw shapes in 3D space. For that humans manually manipulated the soft arm to follow shapes and recorded the 3D tip trajectory as expert demos. Also prior to policy training, they train a long short-term memory (LSTM) [112] based dynamic behavioural map (DBM), which predicts pneumatic pressure, needed to reach a state, serving as the expert for Soft DAgger. Similar to DAgger, they collect agent roll-outs in a small buffer in combination with the expert demos to train the agent’s policy  $\pi_\theta$ . For the training step, they propose two different variants. Student policy optimization (SPOT) and policy and model optimization (PMO). SPOT treats the DBM as fixed ground truth data and minimizes

$$L_\theta = \frac{1}{N} \sum ||\tau_\phi - \tau_\theta||^2$$

where

- $\tau_\phi$  is the predicted action from DBM,
- $\tau_\theta$  is the action, performed by the robot arm.



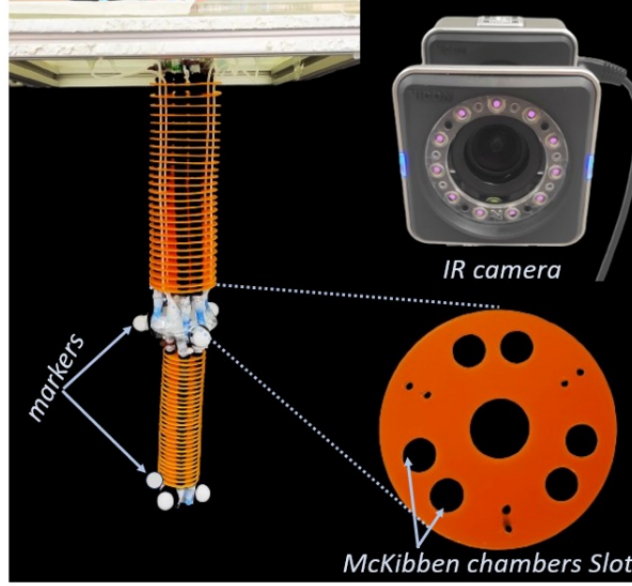


Figure 24: Soft robotic arm with six pneumatic control signals, placed in an eight-IR-camera based motion capture system

PMO jointly refines both DBM and the agent’s policy, by adding a ”behavioural-loss” weighting term for smoother movement

$$L_{\theta} = \frac{1}{N} \sum [\alpha ||\tau_{\phi}^{new} - \tau_{\theta}||^2 + (1 - \alpha) ||\tau_{\phi}^{old} - \tau_{\theta}||^2]$$

where

- $\tau_{\phi}^{new}$  is the latest DBM prediction (after the most recent DBM update),
- $\tau_{\phi}^{old}$  is the DBM prediction for the same state before the last DBM update,
- $\tau_{\theta}$  is the action, performed by the robot arm.
- $\alpha$  weight balancing parameter

Training the DBM and the policy required  $\sim 6 \times 10^3$  state-action pairs, which are  $\sim 99.7\%$  fewer examples, than typical RL-based soft-arm controllers, had 98% accuracy after eight 12.5s trials, performed robust across multiple trajectories, while providing smooth motion with the behavioural-loss weighting.

In a follow up paper from Nazeer et al. [113] presented the first IL-based controller that transfers tropism motions (directional bending responses observed in plants) onto a tendon-driven soft-continuum arm with nine-degrees of freedom (see figure 26. For data collection they recorded ten hours of video data of wheat coleoptiles. They trained the policy with Soft DAgger and were able to reproduce all coleoptile shapes in simulation and on hardware.

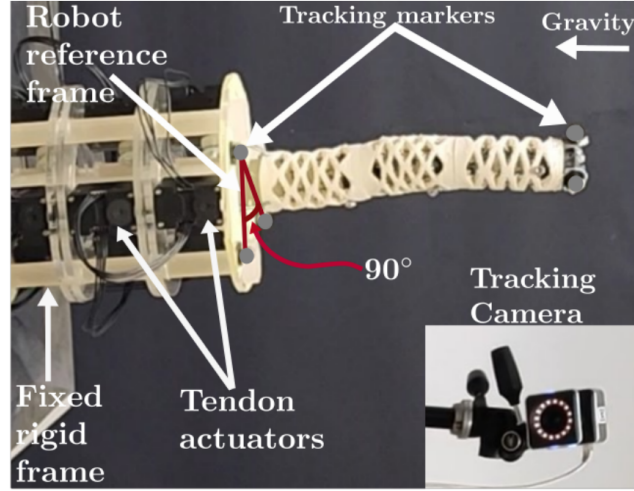


Figure 25: Plant-inspired soft-robotic arm with three independently actuated modules with nine actuators

### 3.9 Rehabilitation Robotics

Rehabilitation robotics leverages assistive and wearable devices technologies to restore or improve motor functions. With recent advances in imitation learning enabling physiologically plausible control of musculoskeletal systems through policy learning.

Humanoid locomotion aims to equip bipedal robots with the ability to move naturally and to transition smoothly between gaits. Compared to wheeled or quadrupedal platforms, humanoids have a high degree-of-freedom, underactuated system. Designing reward terms or analytical models that remain valid across multiple gaits is therefore extremely challenging. Recent work utilizes IL to take advantage of the human motion capture clips and motion priors that implicitly encode natural gait characteristics. Tang et al. [114] introduced HumanMimic, a Wasserstein adversarial imitation (WAIL) framework, that teaches the humanoid JAXON [115] to reproduce a variety of human-like behaviours, including but not limited to standing, squat walking, straight-leg walking and running while being able to switch between on the fly. The HumanMimic contains three parts and is similar to [91] with some changes. The first part handles the skeleton re-targeting that aligns arbitrary human MoCap skeletons with the robot. The second part, velocity-conditioned RL, optimizes a velocity reward, while the third component, a discriminator trained with a soft-boundary Wasserstein-1 loss, returns a style reward. The discriminator’s reward increases for more “human-like” imitation. Training for  $\sim 30$  GPU hours, using Isaac Gym on a mixed dataset yields a single PPO policy that tracks target command velocities up to  $5 \frac{m}{s}$  with smooth gait transitions, absorbs 300 N lateral pushes by performing sidesteps and climbs 50mm stairs in a Chorenoid sim without

additional tuning, which indicates strong generalization.

Simos et al. [116] present KINESIS, the first model-free, muscle-driven controller, that imitates  $\sim 1.9$ h of diverse locomotion clips, follows high level commands, including natural language prompts (using the human motion diffusion model by and produces muscle-activity patterns that line up with real EMG recordings. For their experiments, they use the lower-body MyoLeg model in MuJoCo + MyoSuite with 80 Hill-type muscle-tendon units and 20DoF. The motion capture data was taken from the KIT motion-language database [117]. For training, a MLP policy is optimized using PPO, latent time-correlated exploration and a curriculum based on hard-negative mining, while each curriculum stage yields an "expert" (copy of the policy network that has been trained on a slice of the dataset). A Mixture-of-experts gate then fuses the experts online. KINESIS achieved highly accurate imitation, follows text-generated motions like "walk forward" or "turn left", adapts in real time to moving targets and simulated muscle activations that correlate with human EMG.

Furthermore Dey et al. [118] present a multitask prospective rehearsal (MPR) framework that learns to predict missing limb joint kinematics across diverse gait modes. Their prospective model predicts the next sensor states resulting from the predictor's own output and injects these synthetic states back into training. This effectively performs an offline DAGger without the risk of real-world exploration. Their learning strategy yields state-of-the art prediction accuracy ( $R^2$  up to 0.99) on three large gait datasets. They maintain performance under distribution shift, perturbations and sensor noise, while also preventing catastrophic forgetting for new tasks. In real exoskeleton trials (see figure ??), the joint prediction ran at 50Hz, producing more stable gait and felt naturally synchronizes to users. This marks a significant advance towards robust IL based control for prostheses and wearable rehabilitation devices.

## 4 Conclusion

Imitation learning has progressed from early BC to today's diffusion and transformer-based policies surpassing handcrafted control across diverse domains such as manipulation, locomotion and navigation. This essay systematically shows this progress by examining core algorithms (BC, IRL, AIL, IfO) and their applications in manipulation, autonomous driving, legged and aerial robotics, swarm systems, agriculture, construction and soft robotics. Despite these fundamental advancements, common challenges persist, including covariate shift, multi-modality, data inefficiency, causal misidentification and ensuring safety. However, promising trends such as uncertainty-aware interaction methods, support-based reward mechanisms, multi-modal generative models and enhances sim-to-real strategies are actively bridging this performance gap between robots and humans. Future advances will depend on developing more accurate simulation environments, more efficient and scalable data collection methods, safety frameworks and interpretable and transparent models. This review provides both a comprehensive entry point for newcomers and a roadmap for future research.



Figure 26: Snapshots from the experiment on a subject. In their experiments they tried different motion conditions such as walking at different speeds and inclines on a treadmill (A-E). Ground walking on a level walkway with different motion maneuvers was also tested (F-N)

Ultimately it remains to be said, that further innovation in imitation learning holds the key to transition robotic systems from simulation into dynamical and unpredictable real-world applications.

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